

ISO/IEC JTC 1/SC 42

New Work Item Proposal (NWI)

Compliant with ISO Form 4

A Identification of the Proposal

Proposal Number (To be assigned)	(To be assigned by Secretariat)
Date of Proposal	___ / ___ / 2026
Proposing National Body	Japan (JISC: Japanese Industrial Standards Committee)
Proposer (Originator)	Satoru Hara Independent Researcher, Natural Structure Works (NSW) 1-8-10 Josui-cho, Toyota, Aichi 470-0343, Japan satoru.hara@nifty.com https://www.naturalstructureworks.com/
Submission Route	Via Information Technology Standards Commission of Japan (ITSCJ) Domestic Standards Body for ISO/IEC JTC 1/SC 42

B Title of the Proposed Work

Title (English)	Information technology — Artificial intelligence — Technical Specification for External Verification Architecture (EVA) of AI Systems
Title (French) (For reference)	Technologies de l'information — Intelligence artificielle — Spécification technique pour l'architecture de vérification externe (EVA) des systèmes d'IA
Document Type	International Standard (IS) or Technical Specification (TS) Note: Initial proposal as TS, with transition to IS anticipated following accumulation of implementation track record

C Scope

Scope	This Technical Specification defines the architectural framework — External Verification Architecture (EVA) — for continuously, structurally, and verifiably monitoring, recording, and evaluating the reasoning processes, outputs, and decision-making structures of AI systems through mechanisms independent of the model's internal parameters. Specifically, this document specifies: • The conceptual definition of EVA and its three structural conditions (Verifiability, Transparency, and Physical Immutability) • Implementation requirements for EVA in AI systems (software layer and hardware layer) • Evaluation criteria for EVA conformance and third-party verification procedures • Alignment requirements with existing ISO/IEC AI standards (ISO/IEC 42001, ISO/IEC 23894, etc.) • Recommended architectural patterns for EVA implementation The scope of this specification covers all AI systems including high-risk AI systems, with intended application to industrial, medical, financial, governmental, and infrastructure sectors.
Exclusions	The following are outside the scope of this specification: • AI system capability evaluation and performance benchmarking (addressed by ISO/IEC 29119 etc.) • Process certification for AI management systems (addressed by ISO/IEC 42001) • Specification of particular AI algorithms or implementation technologies

D Purpose and Justification

Background	The safety assurance of current AI systems relies heavily on self-assessment conducted by the developers themselves. International regulations and standards — including the EU AI Act (Regulation 2024/1689), ISO/IEC 42001:2023, and the NIST AI RMF — require 'independent external verification,' yet no concrete architectural framework for technically realizing such verification has been defined as an international standard. The gap between these regulatory requirements and technical implementation is the starting point for this proposal.
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Limitations of Existing Standards	• ISO/IEC 42001: Targets management process certification; external verification of reasoning processes themselves is outside scope • Explainable AI (XAI): Primarily focused on post-hoc interpretation and visualization; not real-time structural external verification • AI audit frameworks: Periodic, snapshot-based evaluation; does not address continuous verification at inference time
Significance of This Proposal	• Establishes the architectural foundation enabling a shift in AI safety assurance from assertion to proof • Functions as the technical specification for EU AI Act Article 12 (log-keeping obligations), enhancing regulatory effectiveness • Ensures cross-industry interoperability through common formats enabling third-party verification • Functions as a basis for mutual recognition of AI products in international trade • Provides the technical infrastructure necessary for a third-party AI safety certification market
Regulatory Alignment	EVA directly addresses the technical implementation requirements of Articles 9, 12, 13, and 17 of the EU AI Act (2024/1689). Coordinated standardization with CEN-CENELEC JTC 21 is envisioned.

E Relevant Existing Documents

ISO/IEC Standards	• ISO/IEC 42001:2023 — Artificial intelligence — Management system • ISO/IEC 23894:2023 — Artificial intelligence — Guidance on risk management • ISO/IEC 22989:2022 — Artificial intelligence — Concepts and terminology • ISO/IEC 23053:2022 — Framework for AI Systems Using Machine Learning • ISO/IEC TR 24028:2020 — Overview of trustworthiness in artificial intelligence • ISO/IEC TR 24029 — Assessment of the robustness of neural networks
Regulations and Frameworks	• EU AI Act (Regulation 2024/1689) — European Artificial Intelligence Regulation • NIST AI RMF 1.0 (2023) — Artificial Intelligence Risk Management Framework • ENISA: Artificial Intelligence Cybersecurity Challenges (2020) • W3C PROV-O: The PROV Ontology Recommendation (2013)

Proposer's Published Documents	Trusted Computing Group: TPM Library Specification, Family 2.0
	The following documents are publicly registered with DOI on Zenodo / SSRN (May 2026): - Hara, S. (2026). Structural Significance and Importance of External Verification Architecture (EVA) in Artificial Intelligence. [Registered on Zenodo / SSRN] - Hara, S. (2026). Human Natural Structure (HNS): A Structural Operating System for Human Understanding in AI. https://github.com/satoru-hara/03_NSW

F Cooperation with Other Bodies

Bodies Requested for Cooperation	- CEN-CENELEC JTC 21: Coordination requested for alignment with EU AI Act technical specifications - IEEE SA (P2863 etc.): Alignment with AI system ethics and transparency standards - ITU-T AI Focus Group: Application to telecommunications and infrastructure AI - NIST (United States): Confirmation of alignment with NIST AI RMF
Liaison Relationships	Liaison with ISO/IEC JTC 1/SC 27 (Information security) is recommended for technical alignment regarding hardware-layer Physical Immutability implementation (FPGA / ASIC / TPM).

G Proposed Project Leader

Proposed Project Leader	Satoru Hara Independent Researcher, Natural Structure Works (NSW) Proposer of External Verification Architecture (EVA) Author of the Human Natural Structure (HNS) Framework <i>Note: The formal Project Leader is understood to be determined through recommendation by JISC / ITSCJ. The proposer expresses willingness to continue contributing as a technical expert.</i>
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H Commitment to Participate

Commitment of Active Participation (P-members)	The following countries / bodies are expected to express commitment to active participation (P-member): • Japan (JISC): Active participation as proposing country • Europe (via CEN-CENELEC): Participation expected from the perspective of EU AI Act alignment • United States (ANSI): Participation expected from the perspective of NIST AI RMF alignment <i>Note: Formal P-member confirmation shall be conducted through the JISC Secretariat</i>
Proposer's Commitment	• Contribution to drafting the first version of the technical specification • Participation in technical review during the Working Draft (WD) stage • Provision of implementation guidance • Provision of related publicly registered technical documents (Zenodo / SSRN)

I Proposed Work Schedule

J Remarks and Additional Information

Prior Technical Documentation	The paper forming the technical foundation of this NWI has been registered with DOI on Zenodo / SSRN (May 2026). It functions as a citable technical basis for the standards proposal.
Relationship Between EVA and HNS	The EVA proposed in this NWI is a conceptual framework and does not depend on any specific implementation technology. HNS is one technical method for realizing EVA, but should be evaluated independently of the content of this standard.
Declaration of Independence	The proposer is an independent researcher with no affiliation or conflict of interest with any company or institution. This proposal aims to establish the public infrastructure for AI safety, entirely separate from any commercial interest.

Signature of Proposer	Endorsement by National Body (JISC)
Signature: _____	Signature: _____
Name: Satoru Hara	Name / Title:
Date: ____ / ____ / 2026	Date: ____ / ____ / 2026